# Chapter 1 — Periodic Table and Periodic Properties

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

## Practice MCQs

**1. Modern periodic law relates properties to:**

**(a)** Atomic mass **(b)** Density **(c)** Atomic number **(d)** Neutron number

**2. Number of periods in the modern table:**

**(a)** 7 **(b)** 10 **(c)** 8 **(d)** 18

**3. Number of groups in the modern table:**

**(a)** 10 **(b)** 18 **(c)** 8 **(d)** 7

**4. Elements in the same group have the same:**

**(a)** Valence electrons **(b)** Number of shells **(c)** Density **(d)** Atomic mass

**5. The last electron enters p subshell in:**

**(a)** s-block **(b)** f-block **(c)** d-block **(d)** p-block

**6. Metalloids lie along the:**

**(a)** Zero group **(b)** Bottom row **(c)** Stair-step line **(d)** Left edge

**7. Across a period, atomic radius generally:**

**(a)** First increases **(b)** Decreases **(c)** Increases **(d)** Remains same

**8. Down a group, atomic radius generally:**

**(a)** Decreases **(b)** Increases **(c)** Becomes zero **(d)** Remains same

**9. A cation is smaller than its parent atom because:**

**(a)** Electrons are lost **(b)** Protons are lost **(c)** Neutrons are gained **(d)** Shells increase

**10. An anion is larger than its parent atom because:**

**(a)** Neutrons are lost **(b)** Nucleus shrinks **(c)** Protons are gained **(d)** Electrons are gained

**11. First ionization energy generally increases:**

**(a)** Across a period **(b)** With shielding **(c)** Down a group **(d)** With size

**12. Second ionization energy of Na is very high because:**

**(a)** Low nuclear charge **(b)** Stable noble-gas core **(c)** Metallic bond **(d)** Large size

**13. Nitrogen has higher IE than oxygen due to:**

**(a)** Lower nuclear charge **(b)** Half-filled p subshell **(c)** Metallic character **(d)** Larger size

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**14. Most negative electron affinity is shown by:**

**(a)** Nitrogen **(b)** Fluorine **(c)** Chlorine **(d)** Oxygen

**15. Most electronegative element is:**

**(a)** Oxygen **(b)** Fluorine **(c)** Chlorine **(d)** Nitrogen

**16. Electronegativity down a group generally:**

**(a)** Decreases **(b)** Increases **(c)** Becomes zero **(d)** Remains same

**17. Sodium with cold water gives:**

**(a)** NaOH and $H_2$ **(b)** NaCl and $H_2$ **(c)** NaH and $O_2$ **(d)** $Na_2O$ and $H_2$

**18. Magnesium reacts with steam to give:**

**(a)** $MgCl_2$ and $H_2$ **(b)** MgO and $H_2$ **(c)** $MgH_2$ and $O_2$ **(d)** $Mg(OH)_2$ and $O_2$

**19. Sodium reacts vigorously with oxygen in open air to form:**

**(a)** NaO **(b)** $Na_2O$ only **(c)** $NaO_2$ **(d)** $Na_2O_2$

**20. Magnesium burns in oxygen to form:**

**(a)** MgO **(b)** $Mg(OH)_2$ **(c)** $MgO_2$ **(d)** $Mg_2O$

**21. Sodium reacts with chlorine to form:**

**(a)** NaCl **(b)** $NaClO_3$ **(c)** NaClO **(d)** $Na_2Cl$

**22. Across Period 3, metallic character:**

**(a)** Decreases **(b)** First increases **(c)** Remains same **(d)** Increases

**23. $Na_2O$ is:**

**(a)** Basic oxide **(b)** Amphoteric oxide **(c)** Neutral oxide **(d)** Acidic oxide

**24. $Al_2O_3$ is:**

**(a)** Neutral **(b)** Basic **(c)** Acidic **(d)** Amphoteric

**25. $SO_3$ is:**

**(a)** Basic oxide **(b)** Neutral oxide **(c)** Amphoteric oxide **(d)** Acidic oxide

**26. Highest oxidation state in $Cl_2O_7$ is:**

**(a)** +3 **(b)** +7 **(c)** +1 **(d)** +5

**27. NaCl in water gives:**

**(a)** No dissolution **(b)** Acidic solution **(c)** Basic solution **(d)** Neutral solution

**28. $SiCl_4$ in water gives:**

**(a)** Neutral solution **(b)** Basic solution **(c)** Acidic solution **(d)** Precipitate only

**29. $PCl_5$ is:**

**(a)** Network solid **(b)** Metallic **(c)** Ionic **(d)** Covalent

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**30. Period number indicates:**

**(a)** Neutron number **(b)** Valence electrons **(c)** Oxidation state **(d)** Outer principal shell

**31. Dobereiner classified elements into:**

**(a)** Octaves **(b)** Triads **(c)** Periods **(d)** Blocks

**32. Newlands proposed:**

**(a)** Law of octaves **(b)** Modern law **(c)** Atomic number **(d)** Triads

**33. Mendeleev arranged elements by:**

**(a)** Melting point **(b)** Atomic mass **(c)** Atomic number **(d)** Density

**34. Moseley's work established:**

**(a)** Neutron number **(b)** Isotopes **(c)** Atomic number **(d)** Atomic mass

**35. The period number equals:**

**(a)** Neutrons **(b)** Valence electrons **(c)** Oxidation state **(d)** Highest occupied shell

**36. Group 1 elements are:**

**(a)** Noble gases **(b)** Alkali metals **(c)** Halogens **(d)** Alkaline earth metals

**37. Group 2 elements are:**

**(a)** Halogens **(b)** Alkali metals **(c)** Noble gases **(d)** Alkaline earth metals

**38. Group 17 elements are:**

**(a)** Noble gases **(b)** Halogens **(c)** Alkali metals **(d)** Metalloids

**39. Group 18 elements are:**

**(a)** Halogens **(b)** Alkali metals **(c)** Noble gases **(d)** Transition metals

**40. d-block elements are called:**

**(a)** Transition elements **(b)** Noble gases **(c)** Halogens **(d)** Inner transition elements

**41. f-block elements are:**

**(a)** Halogens **(b)** Alkali metals **(c)** Inner transition elements **(d)** Noble gases

**42. Among isoelectronic ions, radius decreases as:**

**(a)** Electrons increase **(b)** Nuclear charge increases **(c)** Mass decreases **(d)** Shells increase

**43. Boron has lower first IE than Be because:**

**(a)** B has fewer protons **(b)** 2p electron is easier to remove **(c)** 2s electron is easier **(d)** B is larger

**44. Electron affinity is energy change when gaseous atoms:**

**(a)** Share electrons **(b)** Lose electrons **(c)** Form bonds **(d)** Gain electrons

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**45. A large electronegativity difference favours:**

**(a)** No bonding **(b)** Ionic bonding **(c)** Metallic bonding **(d)** Non-polar covalent bonding

**46. Sodium is stored under:**

**(a)** Dilute acid **(b)** Water **(c)** Kerosene oil **(d)** Air

**47. MgO is:**

**(a)** Neutral oxide **(b)** Amphoteric oxide **(c)** Acidic oxide **(d)** Basic oxide

**48. $P_4O_{10}$ is:**

**(a)** Neutral oxide **(b)** Amphoteric oxide **(c)** Basic oxide **(d)** Acidic oxide

**49. Across Period 3, oxides change from:**

**(a)** Basic to acidic **(b)** Acidic to basic **(c)** Neutral only **(d)** Amphoteric only

**50. In $Na_2O$, oxidation number of Na is:**

**(a)** 0 **(b)** +2 **(c)** -1 **(d)** +1

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## Answer Key with Brief Reasons

**1. Answer: (c) Atomic number** — Modern law uses atomic number.

**2. Answer: (a) 7** — There are seven horizontal periods.

**3. Answer: (b) 18** — There are eighteen vertical groups.

**4. Answer: (a) Valence electrons** — Group similarity comes from valence electrons.

**5. Answer: (d) p-block** — Block is named by the subshell receiving the last electron.

**6. Answer: (c) Stair-step line** — Metalloids border metals and non-metals.

**7. Answer: (b) Decreases** — Effective nuclear charge increases.

**8. Answer: (b) Increases** — New shells are added.

**9. Answer: (a) Electrons are lost** — Loss of electrons reduces repulsion.

**10. Answer: (d) Electrons are gained** — Added electrons increase repulsion.

**11. Answer: (a) Across a period** — Nuclear attraction increases across a period.

**12. Answer: (b) Stable noble-gas core** — $Na^+$ has neon configuration.

**13. Answer: (b) Half-filled p subshell** — Half-filled p3 is relatively stable.

**14. Answer: (c) Chlorine** — Cl avoids strong repulsion of very small F.

**15. Answer: (b) Fluorine** — Fluorine has highest electronegativity.

**16. Answer: (a) Decreases** — Size and shielding increase.

**17. Answer: (a) NaOH and $H_2$** — $2Na + 2H_2O \rightarrow 2NaOH + H_2$.

**18. Answer: (b) MgO and $H_2$** — $Mg + H_2O \rightarrow MgO + H_2$.

**19. Answer: (d) $Na_2O_2$** — Book gives sodium peroxide in open air.

**20. Answer: (a) MgO** — $2Mg + O_2 \rightarrow 2MgO$.

**21. Answer: (a) NaCl** — $2Na + Cl_2 \rightarrow 2NaCl$.

**22. Answer: (a) Decreases** — Elements become less electropositive.

**23. Answer: (a) Basic oxide** — Metal oxides are basic.

**24. Answer: (d) Amphoteric** — It reacts with acids and bases.

**25. Answer: (d) Acidic oxide** — Non-metal oxide forms acid.

**26. Answer: (b) +7** — O is -2, so Cl is +7.

**27. Answer: (d) Neutral solution** — Ionic chloride of strong base/acid.

**28. Answer: (c) Acidic solution** — Covalent chloride hydrolyses.

**29. Answer: (d) Covalent** — Higher Period 3 chlorides are covalent.

**30. Answer: (d) Outer principal shell** — Period number is the shell being filled.

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**31. Answer: (b) Triads** — Groups of three with similar properties.

**32. Answer: (a) Law of octaves** — Every eighth element repeated properties.

**33. Answer: (b) Atomic mass** — Historical periodic table used atomic mass.

**34. Answer: (c) Atomic number** — X-ray studies gave nuclear charge.

**35. Answer: (d) Highest occupied shell** — Period indicates principal shell.

**36. Answer: (b) Alkali metals** — Group 1 family.

**37. Answer: (d) Alkaline earth metals** — Group 2 family.

**38. Answer: (b) Halogens** — Group 17 family.

**39. Answer: (c) Noble gases** — Group 18 family.

**40. Answer: (a) Transition elements** — Last electron enters d subshell.

**41. Answer: (c) Inner transition elements** — Lanthanoids and actinoids.

**42. Answer: (b) Nuclear charge increases** — Greater charge pulls same electron cloud closer.

**43. Answer: (b) 2p electron is easier to remove** — 2p is higher in energy than 2s.

**44. Answer: (d) Gain electrons** — Definition of electron affinity.

**45. Answer: (b) Ionic bonding** — Electron transfer is favoured.

**46. Answer: (c) Kerosene oil** — Prevents reaction with air/moisture.

**47. Answer: (d) Basic oxide** — Metal oxide reacts with acid.

**48. Answer: (d) Acidic oxide** — Non-metal oxide forms acid.

**49. Answer: (a) Basic to acidic** — Metallic character decreases.

**50. Answer: (d) +1** — O is -2, so Na is +1.

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